Nicolas **Comte**, PhD

Machine Learning Engineer



Profile

Machine Learning Engineer & Computer Science PhD specializing in Deep Learning, computer vision in medical imaging.

Combining expertise in research, engineering, and Agile methodologies, I design and conduct deep learning, modeling, and simulation projects.

Contact details

@ nicolas.comte1@gmail.com

+33 6 75 83 50 60

Page: cometicon.github.io

LinkedIn: nicolascomte

Personal information

Citizenship French

Family Married, 2 children

Languages French (native), English (C1)

Skills

Machine Learning & AI Machine learning, deep learning, computer vision, 2D/3D medical imaging, motion capture

3D & Scientific computing 3D mesh processing, digital twins, simulation

Programming Python, C++, R, Git

Frameworks & Tools PyTorch, Scikit-learn, SOFA, R, Numpy, Pandas, OpenMP, ...

Engineering & Delivery MLOps pipelines, API development, CI/CD, Code reviews, Agile methodology and facilitation

Professional Experience

Machine Learning Engineer, 3D Deep Learning at *Anatoscope–Circle (France).* **2024–2026**

• Architected and deployed scalable deep learning models for automated 3D intraoral scan analysis.
• Guided technical strategy and architecture choices through continuous state-of-the-art monitoring.
• Engineered robust MLOps pipelines to automate model retraining and the end-to-end ML lifecycle.
• Integrated CI/CD workflows and enforced engineering best practices to prevent regressions via rigorous code reviews.
• Facilitated Agile planning, technical scoping, task breakdown, and value-based prioritization.

R&D Engineer & PhD Candidate at *Inria*, *Anatoscope (France).* **2019–2023**

• Conducted industrial research on medical image processing, 3D anatomical modeling, and simulation of scoliotic spines.
• Developed novel machine learning methods integrating motion capture data into 3D structural spine models.
• Published research (Google scholar) and won Best Poster Award in AI for Biomedical Imaging (IABM 2023).

Software Engineer at *Inria (France).* **2016–2018**

• Engineered and implemented fast, open-source C++/Python scientific applications for molecular phylogeny.
See Treerecs and Seaview 5 (> 300 citations).
• Co-supervision of 2 interns (engineering school and BTS vocational degree on Programming)
+ teaching at *INSA Lyon* C++ programming.

Education

PhD in Computer Science. *Université Grenoble-Alpes.* **2020-2023**

• Thesis title: *Learning scoliosis patterns using anatomical models and motion capture.*

Engineer in Bio-Informatics and Modeling. *INSA Lyon.* **2013–2016**

Bachelor in Bio-Informatics. *Université de Lyon.* **2010–2013**

Awards

Best Poster in artificial intelligence applied in biomedical imaging. *IABM 2023, Colloque National en Intelligence Artificielle en Imagerie Biomédicale.*

Gold medal and Best Composite Part. *IGEM 2014, International Genetically Engineered Machine competition.*

Hobbies

Graphics: infographics, illustration, digital art.

Science communication: Animator/speaker (*Geek Touch* 2018, *Fête de la Science* 2017–18), member of *Démesures* 2017–19.

Photography: macrophotography @cometicon.macros

Programming: simulation and artificial life side projects.